

Acute nephritic syndrome in children

*Gemechu Edae (MD, Assistant professor of pediatrics
Dept of Peds, CHS, AU)*

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Objectives

- Understand the pathogenesis and clinical manifestations of A-PSGN and IgA nephropathy
- Understand the approach to a child with nephritic syndrome
- Know the DDX of Nephritic syndrome
- Know the principles of management of APSGN

Acute nephritic syndrome

➤ Sudden onset of:

✓ *Hematuria*

✓ *Oliguria:- < 500 ml/day, or < 0.5 ml/kg/hr)*

✓ *Edema:- Extracellular Vol. expansion*

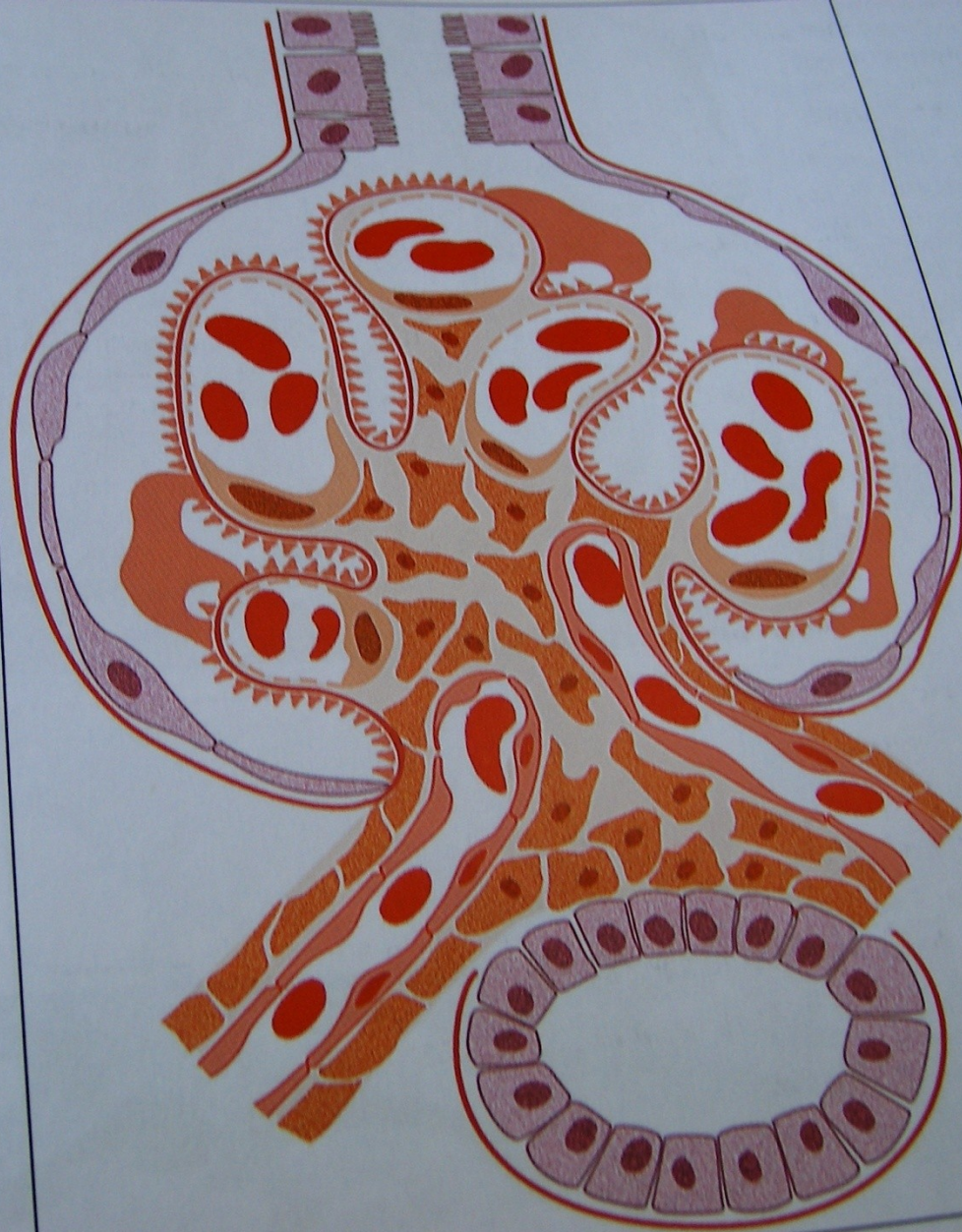
✓ *Hypertension*

✓ *Proteinuria might reach nephrotic range*

➤ **Proliferative GN:-**the classical pathological correlate of AGN

➤ **APGN** - *a prototype of acute nephritic syndrome:*

Fig. 1.1 In the normal glomerulus, the capillary loops are open, the mesangial areas have no more than three nuclei each, and foot processes are intact, without any deposits or proliferation.



Glomerulonephritis

- Is a term reserved for a variety of renal diseases
- The whole mark is inflammation of the glomerulus:-
 - ✓ *Proliferation of cellular elements secondary to an immunologic mechanism*

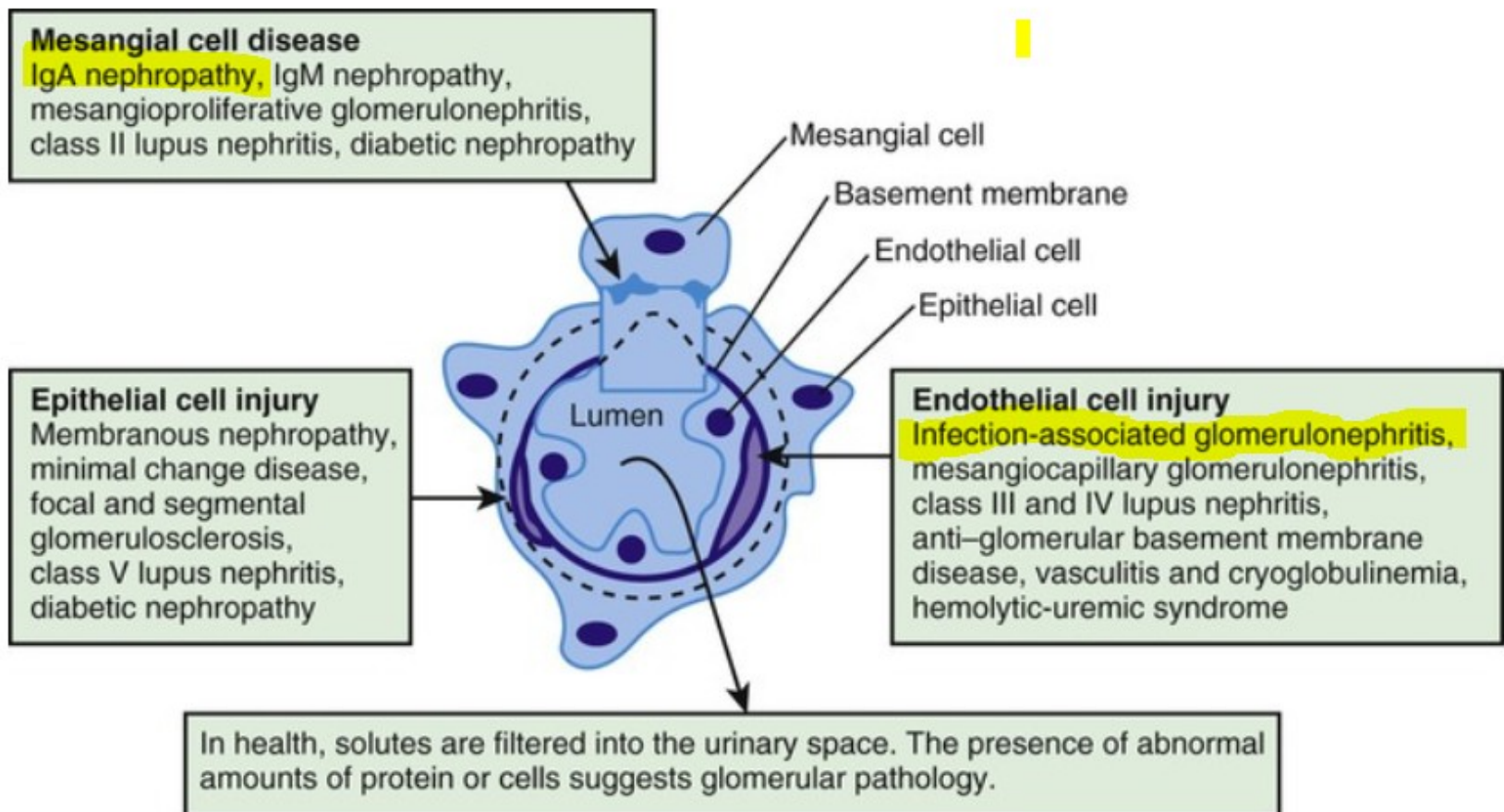


FIG. 535.6 Cellular location of injury during glomerulonephritis. Mesangial cells are directly exposed to the circulation. Deposition of immune complexes within these cells is typically seen in disorders such as immunoglobulin A (IgA) nephropathy; it results in proliferation and expansion of the cells, leading to hematuria, proteinuria, and renal impairment. Epithelial cells, in conjunction with basement membrane, allow filtration of plasma solutes but retard passage of cells and plasma proteins. Disease related to these cells is typified by the presence of subepithelial deposits and flattening of the foot processes that engage the basement membrane, resulting in disruption of the filtration barrier and proteinuria. Endothelial cell disease can result from deposition of immune complex (as

APSGN

➤ Pathogenesis

- The immunologic system may be activated by:
 - ✓ *Streptococcal Ag that adhere to glomerular structures and act as planted antigens or by alterations of endogenous Ag.*
- Ab against streptococci → immune complexes → activation of the complement system → glomerular injury.

Pathogenesis...

- Non-group A streptococci, particularly group C and G have also demonstrated to produce GN.
- Erythrogenic toxin type B produced by:
 - ✓ *Streptococci may be a key part of the immune complexes*
- Pharyngitis related AGN
- Pyoderma related AGN

Pathogenesis...

- The tufts:
 - ✓ Enlarged, swollen,
 - ✓ and a moderate-to-marked increase proliferation of mesangial and epithelial cells is present
- PMNL:- often observed as part of the inflammatory process.

Fig. 1.121 Acute post-infectious glomerulonephritis. There is an exudative proliferation with numerous PMNs and endocapillary proliferation, with scattered mesangial and large hump-shaped subepithelial deposits.

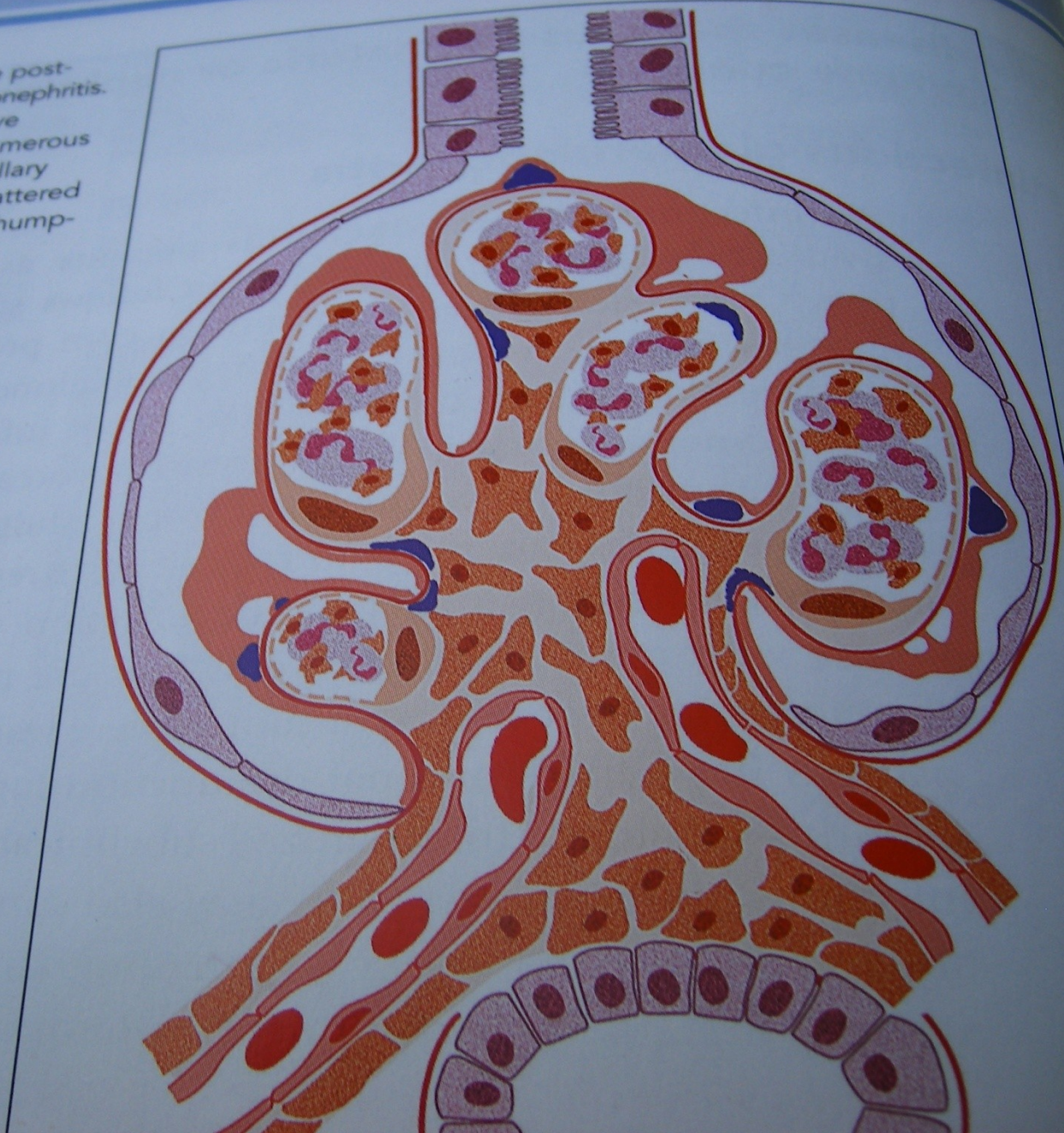


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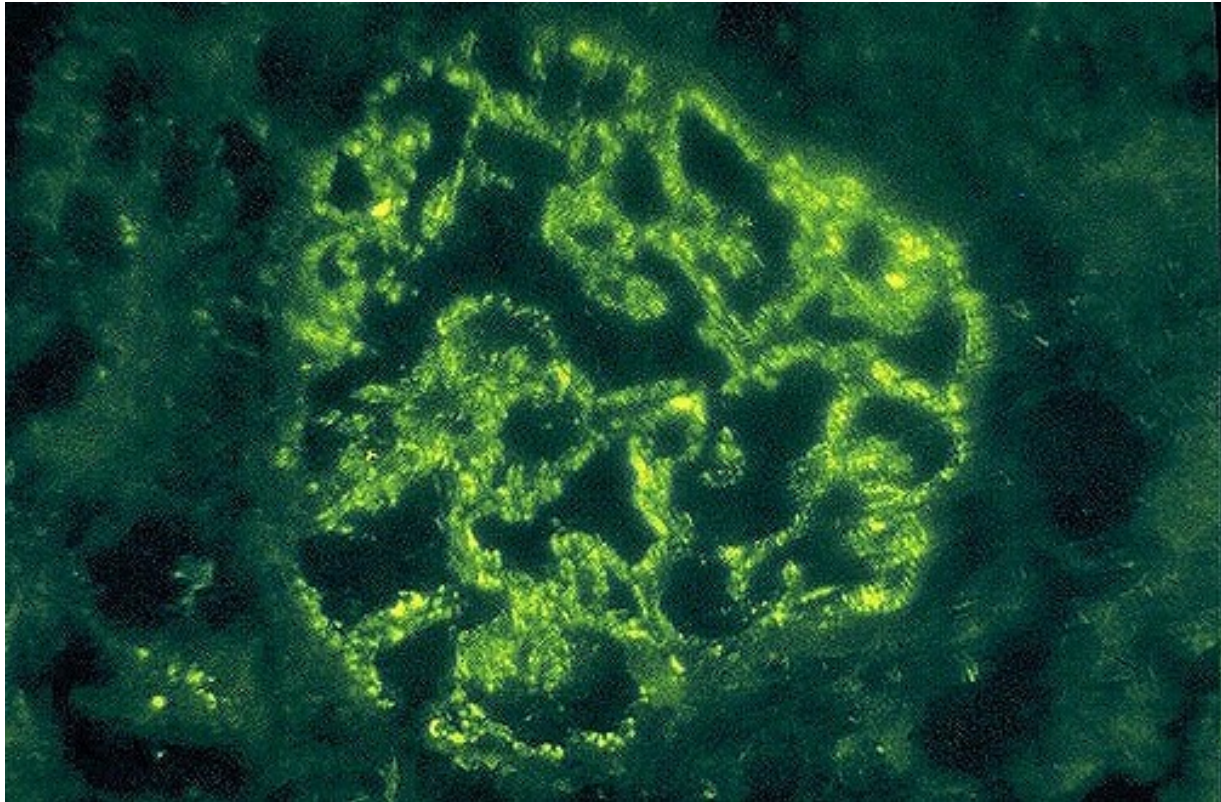
Pathogenesis...

- Reduction in GF leading to:-
 - ✓ Edema formation, HTN, circulatory congestion and encephalopathy
- Various cytokines might be the causes of hypertension
- *Hypertensive encephalopathy*
- Plasma renin level is low, normal or slightly increased

Pathogenesis ...

- **Three** immunofluorescence patterns
 - ✓ Starry sky – due to coarse irregular deposits along the GBM
 - ✓ Garland pattern – due to more elongated deposits along the capillary wall- **worst prognosis**
 - ✓ Mesangial patterns

GRANULAR BUMPY DEPOSITS ON I.F



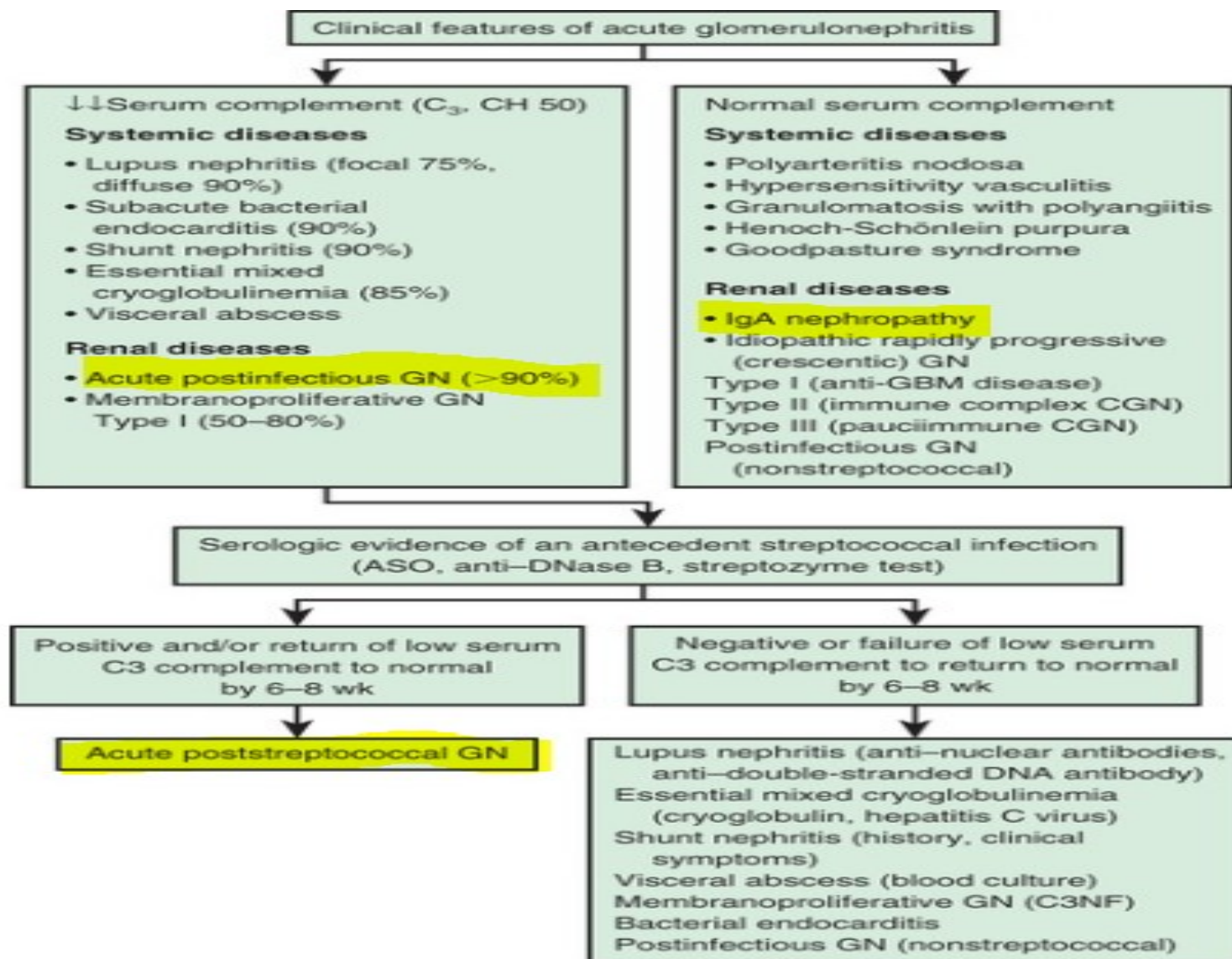


FIG. 537.6 Differential diagnostic algorithm of acute glomerulonephritis (GN). ASO, anti-streptolysin O; GBM, glomerular basement membrane; NF, nuclear factor. (Adapted from Sulyok E: Acute proliferative glomerulonephritis. In Avner ED, Harmon WE, Niaudet P (eds): Pediatric Nephrology, 5/e. Philadelphia, 2004, Lippincott Williams & Wilkins, Fig. 30-4.)

Epidemiology and clinical manifestations

❖ Epidemiology

- Reported as young as 8 months of age,
- Greatest frequency occurs in children 5-12 years
- Uncommon before 3yr.

❖ Clinical

- May be asymptomatic/have sever symptoms
- *Gross hematuria and/or edema: ...*

Clinical manifestations...

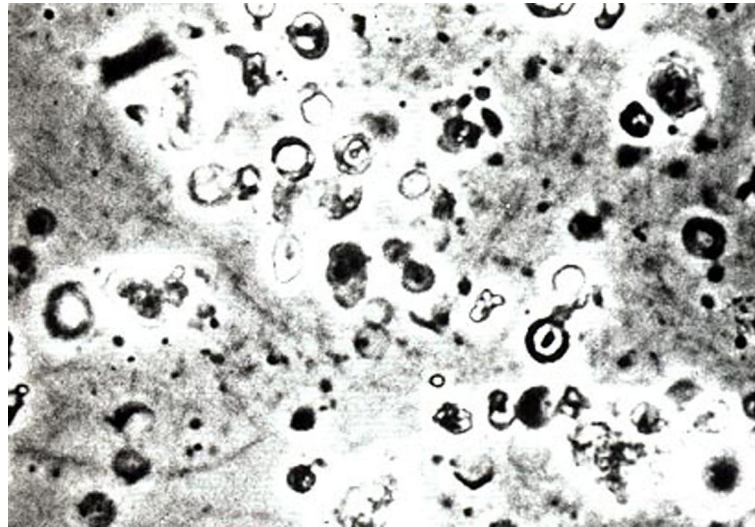
- Insidious onset indicates other forms of renal disease
- Urine usually described as smoky, cola colored,
- Observant parents may note **oliguria**
- **Hypertension is the third cardinal feature, reported**

Laboratory studies

❖ Urine

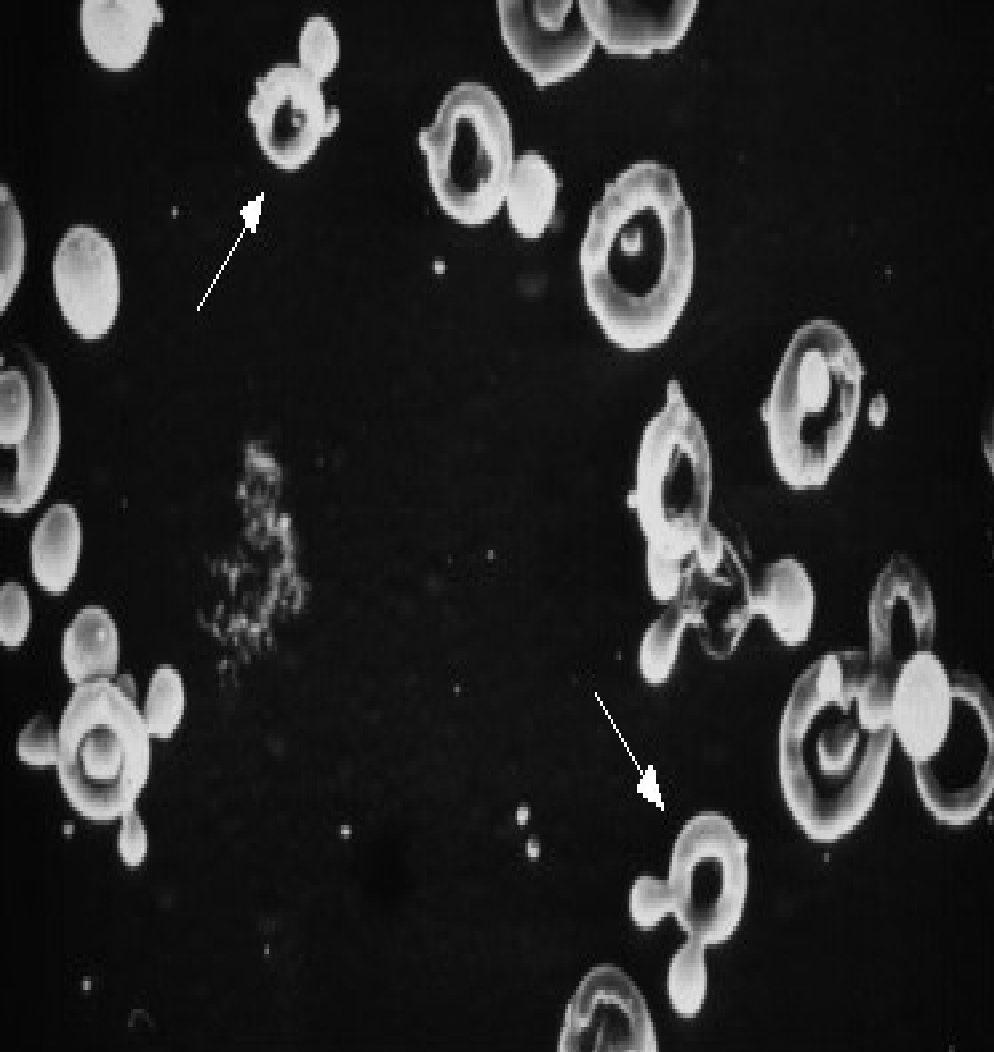
- ✓ Glucosuria occasionally,
- ✓ Proteinuria is commonly present, rarely exceeds 3+
- ✓ *Massive proteinuria and a nephrotic picture*
- ✓ PMNL and renal epithelial cells are common
- ✓ Hyaline and/or cellular casts are almost always present
- ✓ RBC casts in 60-85% of cases

Laboratory – urine sediment

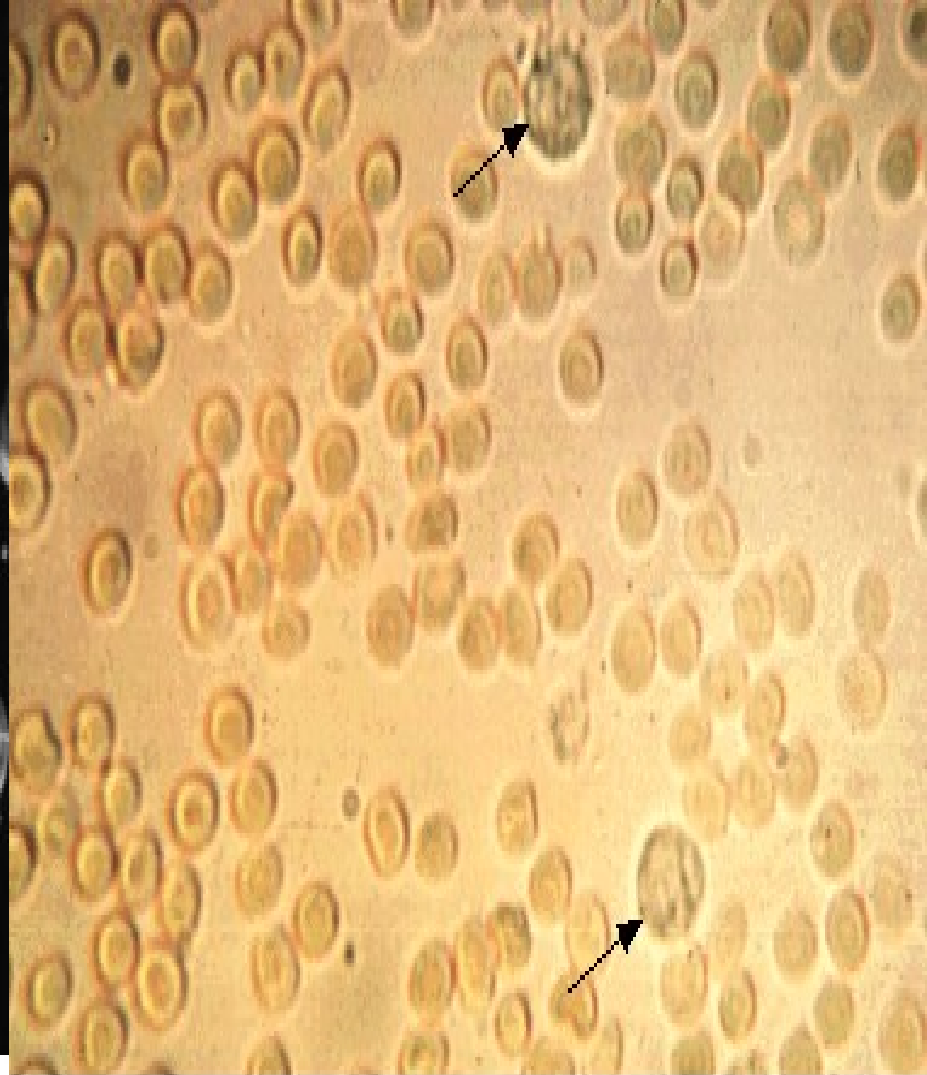


Lab...RBC cast





Dysmorphic red cells Scanning microscopy showing dysmorphic red cells in a patient with glomerular bleeding. Acanthocytes can be recognized as ring forms with vesicle-shaped protrusions (arrows). Courtesy of Hans Köhler, MD.



Monomorphic red cells Urine sediment showing many red cells and an occasional larger white cell with a granular cytoplasm (arrows). The red cells have a uniform size and shape, suggesting that they are of nonglomerular origin. Courtesy of Harvard Medical School.

Lab...

❖ Streptococcal infection

- ✓ Culture from pharynx or throat may be positive
- ✓ Anti-DNase B / ASO titers may be positive

❖ Hemolytic complement

- ✓ Total hemolytic complements are low
- ✓ C3 decreased

Lab ...

- C4 levels are most often normal
- C5 and properdin are usually decreased

❖ Renal

- RF impairment correlates directly to the severity of glomerular injury
 - ✓ Elevated creatinine and BUN in severe azotemia

Lab ...

- Electrolyte profile is usually normal,
 - ✓ $\uparrow K^+$ and metabolic acidosis in severe functional impairment
- Total serum calcium may be low in patients who have NS

❖ Blood

- **A mild anemia** (normocytic, normochromic)
 - ✓ Erythropoiesis may decline in the after month of AGN
- *Hypoproteinemia and hyperlipidemia*

Lab...

❖ Imaging studies

■ Renal U/S :

- ✓ Demonstrates normal to slightly enlarged kidneys with echogenicity

Lab ...

❖ **Renal biopsy should be considered:**

- ✓ Nephrotic syndrome,
- ✓ Absence of evidence of streptococcal infection
- ✓ Normal complement levels
- ✓ When hematuria and proteinuria, diminished renal function, and/or a low C3 level persist more than 2 mo after onset.

Treatment ...

- ❖ Directed at treating the acute effects of
 - ✓ *Renal insufficiency and hypertension.*
- ❖ Decision to admit the child
- ❖ **Control of hypertension**
 - ✓ Mild-to-moderate HTN: Rx with bed rest, fluid restriction
 - ✓ Lasix, nifedipine
 - ✓ ACEIs are effective if no renal impairment
 - ✓ Osmotic diuretics such as mannitol are contraindicated.

Treatment ...

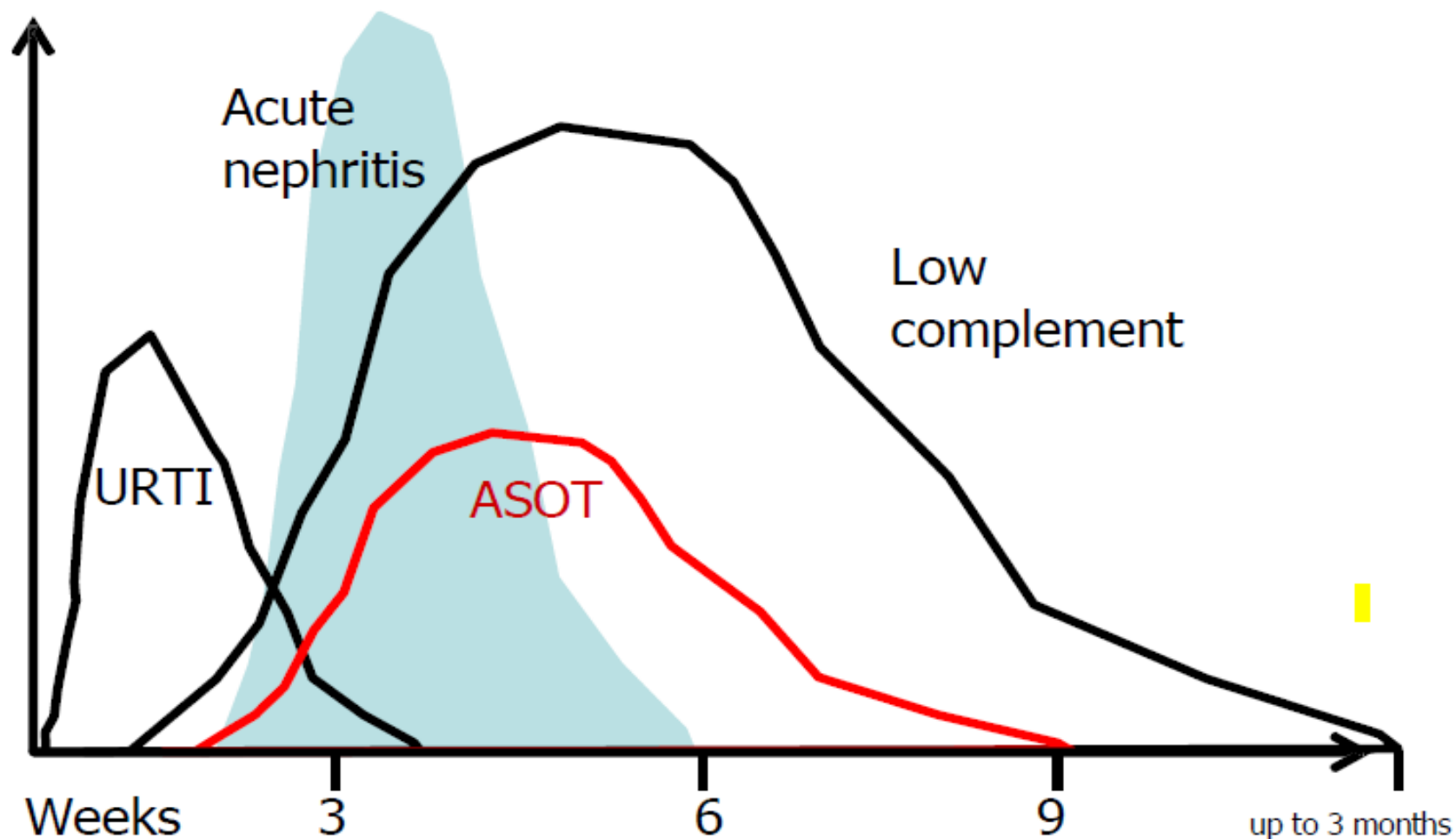
- ❖ A course of **penicillin** administered
- ❖ **Steroid** therapy: severe crescentic GN or in those with RPGN
- ❖ **Diet**
 - ✓ A low Na^+ , low-protein diet during the acute phase
- ❖ **Activity**
 - ✓ Limited activity during the early phase
- ❖ **Follow up**

Prognosis

❖ Prognosis depends on the severity of the initial insult

- ✓ C/F rarely recur after the first 3 months,
- ✓ Second episodes of AGN are uncommon.
- ✓ Edema usually resolves within 5-10 days,
- ✓ HTN & proteinuria usually resolves within 4-6 weeks.

Timing of symptoms, ASOT and complement levels in typical cases of Post-streptococcal GN*



Prognosis...

- Gross hematuria disappears within 1-3 weeks,
- *Microscopic hematuria may persist:*
- C3 returns to normal within 6-8 weeks after onset in 95%.
- Complete recovery occurs in > 95% of children with APSGN.
- CKD in < 2% of affected children

IgA or Berger nephropathy

- Characterized by:
 - ✓ Predominance of IgA deposits within mesangial glomerular in the absence of systemic disease.
- Most common chronic glomerular disease
- More common in male

Causes of IgA Nephropathy

Idiopathic (most cases)

Hepatic cirrhosis

Gluten enteropathy

Minimal change disease

Rare

Membranous nephropathy

HIV infection

Wegener's granulomatosis

Dermatitis herpetiformis

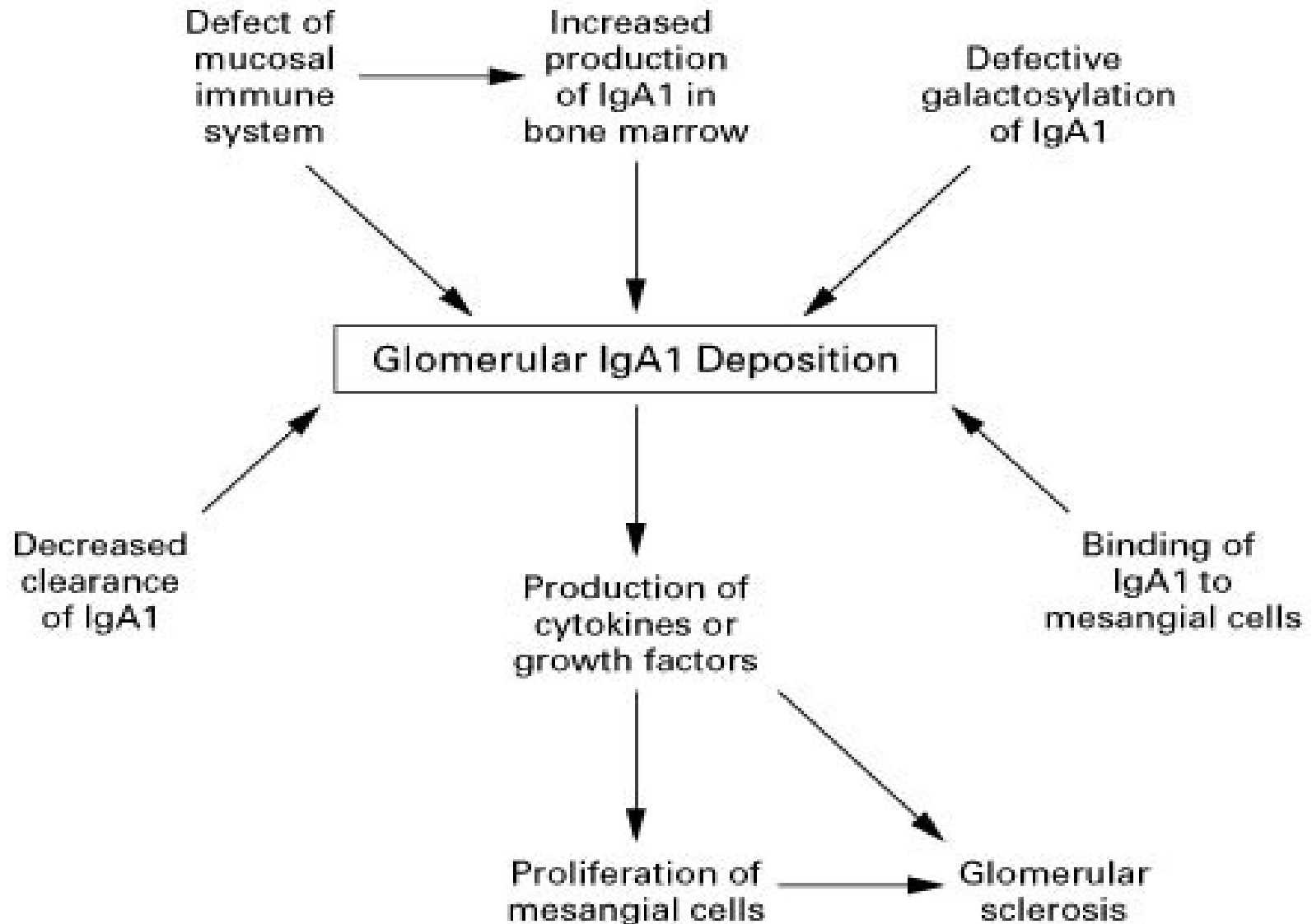
Seronegative arthritis – eg, ankylosing spondylitis

Small cell carcinoma

Disseminated tuberculosis

Mycosis fungoides

Pathogenesis



Pathology

❖ Variable

- Minimal mesangial expansion
- Diffuse proliferative lesion with crescents or widespread sclerosis
- Often mesangial area increase due to increase in both mesangial cells, matrix and deposits.

IgA nephropathy...

❖ Clinical and Lab findings

■ Five main presentations

- ✓ Macroscopic hematuria
- ✓ Asymptomatic microscopic hematuria with or without proteinuria
- ✓ Acute nephritis – hematuria, proteinuria, hypertension, renal failure
- ✓ Nephrotic syndrome (< 10%)
- ✓ Mixed nephrotic/nephritic state

IgA nephropathy...

- Macroscopic hematuria-
 - ✓ Commonest mode of presentation following URTI
 - ✓ *Latent period 1-2 days*
- Elevated serum IgA in 15% of cases
- Serum complement is typically normal
- **Diagnosis:** Renal biopsy

WHO classification of IgA Nephropathy

- I. Minimal lesion
- II. Minor changes with small segmental proliferation
- III. Focal and segmental GN (<50% involved)
- IV. Diffuse mesangial lesion with proliferation and sclerosis
- V. Diffuse sclerosing GN affecting >80% of glomeruli

Treatment

- ❖ Primary treatment is appropriate Bp control
 - ACEIs/ARBs
- ❖ Controversial
 - Fish oil??
 - Steroid??
 - Tonsillectomy??

Prognosis

❖ Worse prognosis:

➤ In older male patients with:

✓ Hypertension,

✓ Marked proteinuria,

✓ ↑Cr at presentation and persistent microscopic hematuria

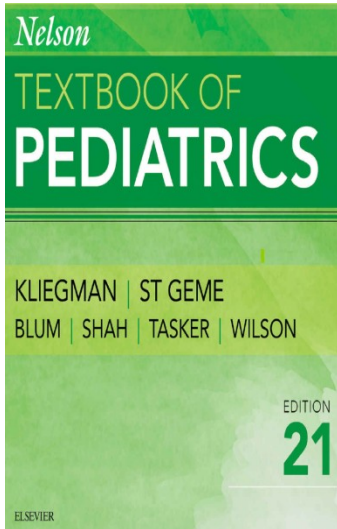
❖ *Progressive disease: 20-30% of patients after 15-20 yrs..*

Table 537.1

Summary of Primary Renal Diseases That Manifest as Acute Glomerulonephritis

DISEASES	POSTSTREPTOCOCCAL GLOMERULONEPHRITIS	IgA NEPHROPATHY	GOODPASTURE SYNDROME	IDIOPATHIC RA PROGRESSIVE GLOMERULONEPHRITIS
CLINICAL MANIFESTATIONS				
Age and sex	All ages, mean 7 yr, 2 : 1 male	10-35 yr, 2 : 1 male	15-30 yr, 6 : 1 male	Adults, 2 : 1 male
Acute nephritic syndrome	90%	50%	90%	90%
Asymptomatic hematuria	Occasionally	50%	Rare	Rare
Nephrotic syndrome	5-10%	Rare	Rare	10-20%
Hypertension	70%	30-50%	Rare	25%
Acute renal failure	50% (transient)	Very rare	50%	60%
Other	Latent period of 1-3 wk	Follows viral syndromes	Pulmonary hemorrhage; iron deficiency anemia	None
Laboratory findings	↑ ASO titers (70%) Positive streptozyme (95%) ↓ C3-C9; normal C1, C4	↑ Serum IgA (50%) IgA in dermal capillaries	Positive anti-GBM antibody	Positive ANCA in
Immunogenetics	HLA-B12, D "EN" (9)*	HLA-Bw 35, DR4 (4)*	HLA-DR2 (16)*	None established
RENAL PATHOLOGY				
Light microscopy	Diffuse proliferation	Focal proliferation	Focal → diffuse proliferation with crescents	Crescentic GN
Immunofluorescence	Granular IgG, C3	Diffuse mesangial IgA	Linear IgG, C3	No immune deposits
Electron microscopy	Subepithelial humps	Mesangial deposits	No deposits	No deposits
Prognosis	95% resolve spontaneously 5% RPGN or slowly progressive	Slow progression in 25-50%	75% stabilize or improve if treated early	75% stabilize or improve if treated early
Treatment	Supportive	Uncertain (options include steroids,	Plasma exchange, steroids,	Steroid pulse therapy

References



University Hospitals of Leicester NHS Trust
Children's Services Medical Guidelines



Thank You

NEPHROTIC SYNDROME

*Gemechu Edae (MD, Assistant professor of pediatrics
Dept of Peds, CHS, AU)*

Outline

- Definitions
- Classification
- Epidemiology
- Pathogenesis
- Clinical features
- Treatment
- Prognosis

Nephrotic Syndrome (NS)

- Definitions
- NS
 - A syndrome characterized by:
 - Proteinuria
 - Hypoalbuminemia ($<2.5\text{g/dl}$)
 - Edema and
 - Hyperlipidemia
 - Nephrotic proteinuria – urinary excretion $> 1 \text{ g/m}^2/\text{day}$ ($40 \text{ mg/m}^2/\text{hr}$) or a first morning protein : creatinine ratio of $>2\text{-}3\text{:}1$

Classification

❖ Etiologic

➤ *Primary/Idiopathic*

- associated with primary glomerular disease

➤ *Secondary*: Occurs as part of recognized systemic disease

- Post infectious (GAS., Syphilis, malaria, TB and viruses)
- Collagen vascular disease (SLE, RA, polyarteritis nodosa)
- Drugs (mercury, gold, penicillin, captopril...)
- HSP, SCD, DM, malignancy (leukemia, lymphoma, wilm's tumor)

Classification...

➤ **Congenital NS**

❖ **Histologic**

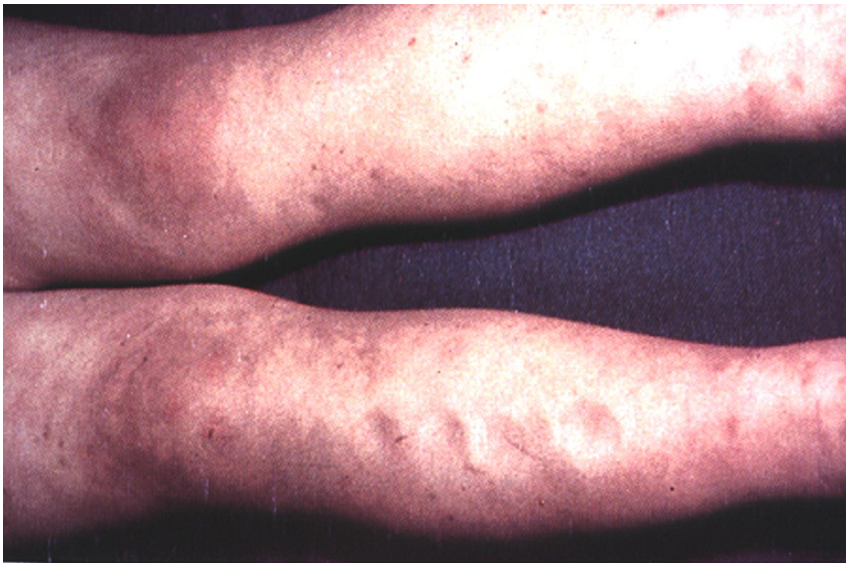
- Minimal lesion NS
- Diffuse mesangial hypercellularity
- Focal glomerulosclerosis
- Membranous glomerulonephritis
- Membranoproliferative glomerulonephritis

Epidemiology

- Incidence 2-7/100,000 children per year
- Primary NS accounts for approximately 90% of all cases
- Age of onset and sex incidence varies with the type of NS
- The most common type of NS in children is **Minimal lesion NS**

Nephrotic syndrome

- Primarily a pediatric disorder
- 15 times more common in children than adults.

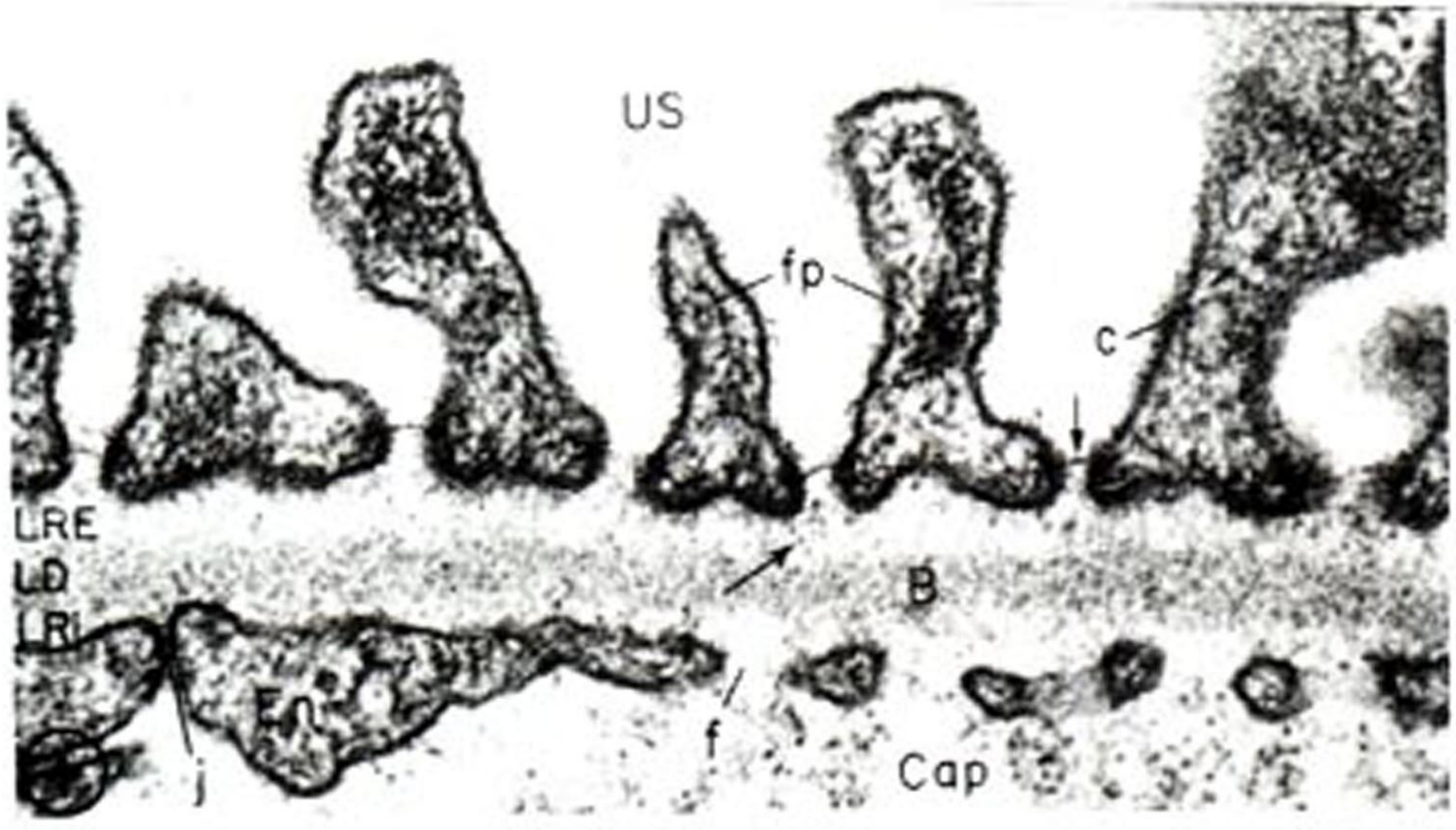


Glomerular handling of proteins

❖ Glomerular capillary wall has three components:

- The fenestrated endothelial cell
- The glomerular basement membrane (GBM)
- The epithelial cell foot processes - the pores between the epithelial foot processes are closed by a thin membrane called the slit diaphragm

Glomerular capillary wall



Glomerular handling.../2

- Filtration barrier by:
 - **Size** (albumin “not allowed”) : the endothelial cell is not part of the size barrier
 - **Charge** (heparan sulphate glycosaminoglycan), the endothelial cell and GBM are negatively charged
 - Slit diaphragm proteins (nephrin, podocin, etc.) Essential for prevention of protein filtration

❖ Mechanisms of **increased protein** excretion:

Glomerular proteinuria

- Commonest cause in children
- Due to increased filtration of macromolecules like albumin
- May be due to anatomic or functional lesions
- The cause might be fever, exercise, or orthostatic

Tubular proteinuria:

- Due to reduced reabsorption of low molecular weight proteins in the proximal tubule
- Beta-2 and alpha-1 microglobulins and retinol binding protein are excreted
- usually associated with the Fanconi syndrome, glycosuria, phosphaturia, PRTA, cystinuria...or tubulointerstitial nephritis

Overflow proteinuria

- Due to increased production of a particular LMWP to a level beyond the absorptive capacity of the proximal tubule.
- Increased production of Ig light chains in multiple myeloma, usually in adults is an example

Idiopathic (Minimal Lesion) NS

- Accounts for approximately 90% of nephrosis in childhood
- It occurs in three morphologic patterns:
 - Minimal change disease (85%)
 - Focal segmental glomerulosclerosis (10%)
 - Mesangial proliferative (5%)
- Occurs between 2-6 yrs commonly
 - But is reported as early as 6 months and throughout adulthood
- Male to female ratio is ~ 2:1

Idiopathic...

- **Pathphysiology**

- Results from alteration in the permeability selective characteristics of the GBM
- Protein loss is generally $> 2\text{g}/24\text{ hr}$ (normal $<150\text{ mg}/24\text{ hr}$)
- Protein lost is primarily albumin resulting in hypoalbuminemia
- Edema appears when serum albumin falls below 2.5 g/dl
- Normal protein excretion in children $< 100\text{ mg}/\text{m}^2/\text{day}$ or $<4\text{ mg}/\text{m}^2/\text{hr.}$ or a total of 150 mg/d , in the neonates up to 300 mg/d

- Pathophysiology.....

The **normal low rate** of protein excretion is due to:

- Restriction of filtration of proteins across the GBM
- Reabsorption of the filtered proteins by the proximal tubule

Idiopathic...

- Serum level of almost all lipids is increased as a result of increases in lipoprotein synthesis and low levels of lipoprotein lipase
- Clinical Manifestations
 - Attacks may follow URTI
 - Usually presents
 - With edema, initially noted in periorbital area and lower extremities
 - Edema becomes generalized with time
 - Some may present with hypotension

Atypical features

- Infants and children >10yrs
- Positive family history of NS
- Presence of extra renal symptoms like rash, arthritis, anemia
- Hypertension
- Renal insufficiency
- Gross hematuria

Idiopathic...

- Rarely may develop renal failure as a result of the hypotension
- With massive pleural effusion and ascites, patients may develop dyspnea
- Unlike nephritic syndrome, gross hematuria and hypertension are uncommon

Idiopathic...

➤ Lab findings

- Urinalysis reveals proteinuria of +3 or +4 on dipstick
- Serum albumin - <2.5 g/dl
- High serum cholesterol and triglyceride level
- Normal C3 level
- Renal biopsy - is indicated in children with:
 - Gross hematuria, HTN, renal insufficiency, hypocomplementemia and whose age is <1 yr or >8yr

Complications

❖ Infection

- Commonest infection is spontaneous peritonitis
 - Commonest cause – strep. Pneumoniae
- Other infections include
 - Pneumonia, cellulitis, sepsis and UTI

❖ Hypercoagulability

❖ ARF

❖ Malnutrition

DDx for edematous child

- Protein losing enteropathy
- Hepatic failure
- Heart failure
- SAM
- Acute or chronic GN

Idiopathic...

➤Treatment

1.General

- Children with mild to moderate edema can be managed as outpatient
- Salt restriction
- High protein diet
- Diuretics can be used cautiously to reduce severe form of edema:
 - Frusemide
 - Albumin
- Careful monitoring of RF and electrolytes

Idiopathic...

2. Steroids

■ Consideration:..

- Prednisolone – given in a dose of $60 \text{ mg/m}^2 / 24 \text{ hr}$ or $1\text{-}2 \text{ mg/kg/24 hr}$ (Max. 80 mg/24 hr) in 3 or 4 divided dose for at least 4 wks
 - 80-90% of patients respond in a median time of 10 days to remission
- After the initial 4-6 wk course, Prednisolone should be tapered to $40 \text{ mg/m}^2/24 \text{ hr}$ every other day every morning

Idiopathic...

- The alternate day dose is slowly tapered over the next 2-3 months
- Response to steroids
 - >95% with minimal change NS
 - ~50% with mesangial proliferation NS
 - ~20% with focal glomerulosclerosis NS
- Cytotoxic drugs
 - Cyclophosphamide – used in frequently relapsing NS or in severe steroid toxicity or in steroid resistant NS

Definitions...

- Steroid responsive NS
 - Disappearance of the proteinuria within 8 wks of starting treatment with a corticosteroid
- Steroid resistance NS
 - When there is no remission after 8 wks of treatment with steroid
- Steroid dependent NS
 - When a patient is unable to discontinue corticosteroid use without experiencing a relapse

Definitions...

- Remission of NS
 - When urine is negative or trace for protein for a period of at least 3 days
- Relapse of NS
 - The occurrence of proteinuria measuring 2+ or more on 3 consecutive days in a patient who is in remission
- Frequently relapsing course
 - Greater than 2 relapse of NS with in 6 months

Idiopathic...

- Prognosis
 - Most children with steroid responsive NS have repeated relapses till the end of 2nd decade

THANK YOU

• ANY QUESTIONS ?

UTI

*Gemechu Edae (MD, Assistant professor of
pediatrics*

Dep't of Peds, CHS, AU)

April 05, 2022

Learning objective

- List etiology and prevalence of UTI
- Classify UTI
- Explain pathogenesis of UTI
- List risk factor of UTI
- Outline diagnosis and treatment of child with UTI

UTI

➤ **Infection in any part of urinary system**

➤ Prevalence and Etiology

- ❖ Occur in 4-7% of infants & 3-5% of school age children with **FEVER**
- ❖ 3-5% of girls and 1% of boys affected
- ❖ In girls average age at first diagnosis – 3 yrs.
- ❖ In boys – first yr of life
- ❖ First year – M:F ratio is 2.8:5.4
- ❖ Beyond 2 yr – M:F ratio is 1:10
- ❖ Mainly caused by **colonic bacteria**

UTI...

- In females 75-90% caused by *E. coli* followed by *Klebsiella* and *proteus*
- In males > 1 yr *proteus* is as common as *E. coli*
- Other causes are viral (adenovirus) – especially in cystitis

➤ Classification

- ✓ Complicated vs. uncomplicated;
- ✓ Three basic forms:

UTI...

❑ Pyelonephritis

❖ Manifests with any or all of the following:

- Abdominal or *flank pain, fever, malaise, vomiting*, jaundice in neonates
- Some newborns may show non specific symptoms such as poor feeding, irritability and weight loss

❖ Acute form may result in renal scarring

❑ Cystitis

❖ Indicate bladder involvement, and manifest:

- With dysuria, urgency, frequency, supra-pubic pain, incontinence and malodorous urine

UTI...

❖ Asymptomatic Bacteriuria

- A positive urine culture with out any manifestation of infection
- Occurs exclusively in girls
- Benign and does not cause renal injury
- *Rx:..*

❖ Pathogenesis

- Ascending infection from fecal flora
- Rarely hematogenous
- Infected urine gets to intra-renal area → inflammatory response → renal injury and scarring

UTI...

• Risk Factors

✓ Female ✓ Uncircumcised male ✓ Vesico-urethral reflux ✓ Toilet training ✓ Voiding dysfunction ✓ Obstructive uropathy ✓ Urethral instrumentation ✓ Wiping from back to front ✓ Pregnancy	✓ Tight underwear ✓ Pinworm infestation ✓ Constipation ✓ Anatomic abnormalities ✓ Neurogenic bladder ✓ Sexual activities
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UTI...

❖ Diagnosis :

➤ Urine culture; depends on having proper urine sample:

- In toilet trained children
 - Mid-stream urine
 - If $> 100,000$ colonies of a single pathogen or
 - If 10,000 colonies and child is symptomatic ~ UTI
 - In uncircumcised children prepuce must be retracted

UTI

In infants

- Sterile **collection bag** after skin disinfection may help to avoid contamination
- Symptoms + > 100,000 colonies ~ Presumed UTI
- If above is not possible, catheterization needed for urine collection

■ Urinalysis

- Pyuria (leucocytes in urine) suggests infection, but infection can occur with out pyuria or pyuria can occur with out infection

Interpretation of culture results in children:

- ✓ single organism $>10^6$ cfu/L (10^3 cfu/mL) may indicate UTI in voided urine
- ✓ any growth from a supra-pubic aspirate is significant
- ✓ pyuria $>10^7$ WBC/L (10^4 WBC/mL) usually indicate UTI, especially with clinical symptoms but may be absent

UTI...

- Positive Nitrites and leukocyte esterase
- Microscopic hematuria (especially in cystitis)
- WBC casts

❖ Hematology

- Leukocytosis, neutrophilia, elevated ESR and CRP are common

❖ Treatment

- Acute cystitis
 - Trimethoprim – Sulfamethoxazole or
 - Nitrofurantoin or
 - Amoxicillin for 5 days

UTI...

❖ AFI suggestive of Pyelonephritis

- Ceftriaxone or
- Ampicillin + Aminoglycoside for 10-14 days



Thank You